HBSM: Hybrid Boundary Sensitivity Mechanisms

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Abstract

This is a living paper shell for a preregistered Mac-first experiment. HBSM tests whether sensitivity in frontier hybrid reasoners can be explained and predicted cheaply. The broad forward-only sensitivity claim is wounded by KL Lens-style forward-sensitivity work, so no novelty or positive result is claimed yet.

1 Current Gate

The branch must pass sensitivity heterogeneity, no-forward predictor, and mechanism gates. If it overlaps with HORN, it should be folded into a unified hybrid-quantization paper. B1 must first reproduce sensitivity heterogeneity on current hybrid reasoners, B2 must beat no-forward baselines without overfitting depth or size, and B3 must add mechanism beyond HORN's boundary-direction test.

2 Mac Artifact Status

A deterministic synthetic B1 real-schema rehearsal now validates the layer-sensitivity packet contract. It emits 504 rows, including 480 primary prompt rows and 24 control rows, and aggregates those primary rows to 40 scoring layers. The packet passes the real checker in non-promoting mode with `SCHEMA_REHEARSAL_NOT_PROMOTABLE_SYNTHETIC_HBSM_B1`. This is synthetic-only: it is not model evidence, not a novelty claim, and not GPU evidence. The evaluator derives measured top-decile membership from aggregated `kl_or_nll_drift`; supplied `top_decile_flag` values are accepted only when they match that measured ranking for every prompt row inside each measured layer. The metadata-only model eligibility packet identifies the current live hybrid targets and architecture hashes, but no corresponding weights are cached repo-locally. HBSM therefore has no real forward-sensitivity evidence on this Mac yet.

3 Next Evidence

The next admissible row is a real layer-sensitivity sweep on current hybrid reasoners. If cheap predictors fail to correlate with forward sensitivity, HBSM should be killed or folded into HORN rather than kept as a separate paper.

4 Admissible Real Packet

The real B1 packet must use fixed boundary flags from the shared architecture maps. Each row must report `model_id`, `layer`, `boundary_flag`, `precision_perturbation`, `kl_or_nll_drift`, `cheap_predictor`, `parameter_count`, `weight_norm`, `top_decile_flag`, `random_top_decile`, `train_test_split`, and `control_type`. The packet must include `config.json`, `raw_rows.jsonl`, `summary.json`, `summary.md`, `decision.md`, `prompt_ids_hash`, `architecture_map_hash`, `trace_plan_hash`, matching `row_count`, and pass `experimental.shared.check_gate_packet --mode real --project hbsm`. The current capture-manifest templates are generated by `experimental.shared.hybrid_trace_capture_manifest`; they must be filled from real forward-sensitivity captures before packet building. The checker requires both true and false boundary flags, finite sensitivity and predictor fields, all listed controls, and a perturbation-off row whose drift is near zero. It also requires exact measured top-decile cardinality $\lceil 0.10 \cdot \text{scoring layers} \rceil$, where scoring aggregates primary prompt rows to one row per `(model_id, layer)`, agreement between supplied top-decile labels and measured drift ranking on every prompt row, an equally sized random baseline that is not enriched, KL-style and activation/outlier comparator controls, and both train and test split rows unless the packet records an explicit resource limit. Any resource-limited packet must use a `RESOURCE_LIMITED_NOT_PROMOTABLE` decision. The summary must include primary-row, scoring-layer, and prompt counts; top-decile count; random top-decile count; train/test counts; boundary top-decile enrichment; Fisher p-value; cheap-predictor Spearman correlation; baseline Spearman values; and the cheap-predictor margin over the best baseline. The checker recomputes these fields from `raw_rows.jsonl` and rejects stale or hand-filled `summary.json` entries.

Gate	Promotion threshold
B1 sensitivity	Primary boundary-only rows show top-decile enrichment over non-boundary rows with Fisher $p < 0.05$, while matched random flags do not reproduce the enrichment.
B2 predictor	Spearman $\rho \geq 0.6$ with bootstrap lower bound ≥ 0.3.
B3 mechanism	Matched-noise attention drift exceeds SSM drift and aligns with HORN.

Table 1: Pre-registered HBSM gate thresholds before any standalone claim.

Required baseline	Purpose
Random top-decile flags	Tests enrichment beyond chance.
Layer-index baseline	Controls depth effects.
Parameter-count baseline	Controls layer size.
Weight-norm baseline	Controls simple magnitude predictors.
Boundary-only baseline	Tests whether HORN already explains the effect.
KL-style ranking baseline	Separates HBSM from forward-sensitivity ranking.
Activation/outlier baseline	Separates boundary sensitivity from generic outlier quantization.
Perturbation-off row	Catches hook and scoring bugs.
Train/test split	Reduces cheap-predictor overfit.

Table 2: Controls required before HBSM can claim a cheaper sensitivity predictor.

5 Reviewer Packet

The reviewer-facing claim boundary is tracked in `experimental/hbsm/paper/reviewer_pack.md`. It records that HBSM is not camera-ready as a standalone paper until B1–B3 separate the mechanism from existing sensitivity tools.

6 Novelty Boundary

The closest risk is KL Lens-style forward sensitivity for mixed-precision hybrid models KL Lens (2026). HBSM can survive only if it shows a cheaper no-forward predictor or a boundary mechanism on current hybrid reasoners; otherwise it should be folded into HORN or killed.

References

Jason Kong, Nilesh Prasad Pandey, Flavio Ponzina, and Tajana Rosing. A KL Lens on Quantization: Fast, Forward-Only Sensitivity for Mixed-Precision SSM-Transformer Models. arXiv:2604.13440, 2026. https://arxiv.org/abs/2604.13440.